

SUDHARSAN S

AI ENGINEER

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SUMMARY

AI Engineer skilled in Python, TensorFlow, PyTorch, and OpenCV with hands-on experience developing real-time AI and computer vision applications. Built projects including driver drowsiness detection, sign language translation, and embedded AI systems using deep learning and Raspberry Pi. Strong understanding of machine learning, data preprocessing, model training, and real-time inference with a focus on practical AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Machine Learning: Scikit-learn, OpenCV

Deep Learning: YOLO, TensorFlow, Keras, PyTorch

Data Analysis: Pandas, NumPy, Matplotlib

Tools & Platforms: GitHub, n8n, VS Code

Embedded & IoT: Raspberry Pi, Arduino

PROJECT EXPERIENCE

Reviva: Real-Time Driver Drowsiness Detection

Jan 2026 – Apr 2026

Python, YOLO, OpenCV, Raspberry Pi

[GitHub Repo](#)

- Developed a real-time driver monitoring system using YOLO and OpenCV for drowsiness detection.
- Integrated Raspberry Pi-based buzzer alerts and Telegram notifications for instant fatigue warnings.
- Optimized video processing pipeline for low-latency real-time detection.

DubSync: Multilingual Video Dubbing with Voice Cloning

Oct 2025 – Dec 2025

Python, Whisper, F5-TTS, Transformers

[GitHub Repo](#)

- Developed an AI-powered video dubbing system with speech transcription, translation, and voice cloning.
- Integrated Whisper ASR and F5-TTS to generate synchronized multilingual dubbed audio.
- Improved speech timing alignment for natural and realistic video dubbing.

SignSense: Real-Time Sign Language Translator

Aug 2024 – Nov 2024

Python, MediaPipe, TensorFlow, OpenCV

[GitHub Repo](#)

- Developed a real-time sign language recognition system using MediaPipe and deep learning.
- Converted hand gestures into English text for communication accessibility.
- Processed live webcam input for real-time gesture tracking and prediction.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning

Nov 2022 – Present

Rajalakshmi Engineering College, Chennai

CGPA: 7.75

CERTIFICATIONS

- Prompt Design in Vertex AI — Google Cloud** Jan 2026
Prompt engineering, LLM workflows, generative AI, Vertex AI, responsible AI
- Building Computer Vision Applications with Python** LinkedIn Learning | Apr 2025
Hands-on image processing projects using Python and OpenCV

PUBLICATION

- Co-authored “DubSync: Deep Learning-Based Cross-Language Video Translation With Voice Cloning” submitted to the ICISCN 2026 IEEE Conference.